

DBCLS BioHackathon 2026 report: QPX: Pathway analysis environment for non-model organisms

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Abstract

We report progress on Quest for Pathways with eXpression (QPX), a pathway analysis environment for non-model organisms, developed during the DBCLS BioHackathon 2026 (BH26). Building on prior BH23/BH25 work, we demonstrated PathLift, a tool that lifts WikiPathways GPML pathways to other species via ortholog mapping, using an Arabidopsis-to-rice mitochondrial complex I example. We also created annotated pathway maps in QPX integrating transcriptome/proteome data for hypoxic stress response, and transcriptome/metabolome data for the non-model organism *Symplocarpus renifolius*, which lacks an established gene ID system. This clarified data-formatting requirements for linking QPX maps to expression tables and surfaced a string-identifier handling issue in QPX's backend, reported upstream. Future work includes reproducible ChEBI ID assignment, automated pathway digitization, and compound-centered functional networks from PubChem/PubTator3 data.

Introduction

As part of the DBCLS BioHackathon 2026, we here report the development of pathway analysis environment called Quest for Pathways with eXpression (QPX) designed for non-model organisms.

Results and Discussion

Following BH23 (Pico et al., [2023](#)) and BH25 (Bono et al., [2026](#)), discussions continued at BH26 with the aim of enhancing the usability of QPX. At BH26, together with three new members from Hiroshima University (two undergraduate students and one postdoctoral researcher), we worked on the following points toward the development of pathway data for non-model organisms.

PathLift in use

PathLift is a command line interface (CLI) tool that 'lifts' pathways in Graphical Pathway Markup Language (GPML) from WikiPathways by mapping them to gene IDs in other species, using an ortholog mapping table. It outputs a GPML file in which the GeneProduct nodes from the source species (e.g. human) have been replaced and expanded to candidate gene nodes for the target species. Create new pathways for non-model organisms with public DB ID annotations

Following domestic BioHackathon in Japan (BH26.6), we conducted several projects to generate pathway data for non-model organisms using PathLift.

As an example of cross-species pathway reuse, we used PathLift to map an *Arabidopsis thaliana* mitochondrial complex I pathway to *Oryza sativa* gene candidates (Fig. 1). The resulting GPML retains the pathway layout while displaying the corresponding rice gene IDs.

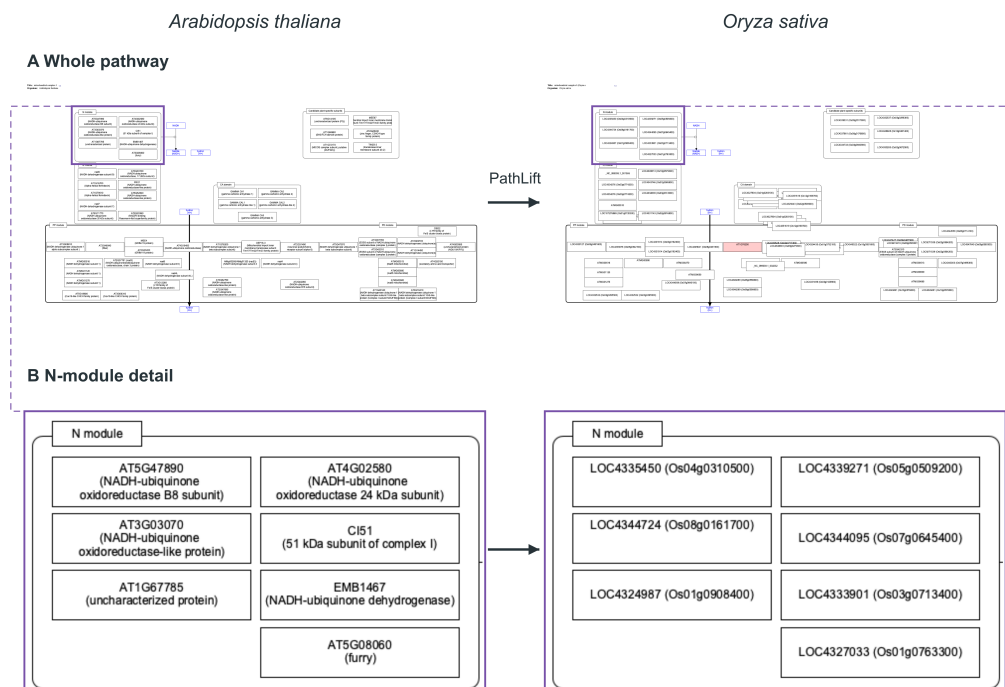


Figure 1: Arabidopsis-to-rice pathway liftover: whole pathway and N-module detail

Figure 1. Arabidopsis-to-rice pathway liftover using PathLift. (A) Whole-pathway views before (left) and after (right) mapping to rice gene candidates. (B) The boxed N-module regions are enlarged to show the gene-label changes.

QPX in Practice

We created several annotated pathway diagrams and used them for biological pathway analysis (Table 1).

Table 1: Newly created use cases in BH26

Pathway	Species	Note
Glycolysis and gluconeogenesis (WP534)	<i>Homo sapiens</i>	Integration of (meta-analysis of) transcriptomes and proteome data
Pentose phosphate, etc	<i>Symplocarpus renifolius</i>	Integration of metabolome and transcriptome data
Food Allergy-Associated Mast Cell Activation Pathway	<i>Homo sapiens</i>	
Mitochondrial complex I	<i>Arabidopsis thaliana</i> <i>Oryza sativa</i>	
Photosynthetic carbon reduction (WP1461)	<i>Arabidopsis thaliana</i>	

Integration of transcriptomes and proteome data

Use case for the integration of multiple omics data was investigated. Differentially abundant proteins (DAPs) in hypoxic stress from proteome and Meta-analysis of public transcriptomes and hypoxic stress were extracted from manuscripts below.

1. Meta-analysis of public transcriptomes: hypoxic stress RNA-seq data (HN-score) collected in “Multi-Omic Meta-Analysis of Transcriptomes and the Bibliome Uncovers Novel Hypoxia-Inducible Genes (Ono & Bono, 2021).
2. Proteome: differentially abundant proteins (DAPs) flags in “Proteomic-Based Analysis of Hypoxia- and Physioxia-Responsive Proteins and Pathways in Diffuse Large B-Cell Lymphoma (Duś-Szachniewicz et al., 2021).

Integration of transcriptome and metabolome data

As a use case of QPX for a non-model organism, we prepared pathway maps together with gene expression and metabolome data of the thermogenic spadix of Asian skunk cabbage (*Symplocarpus renifolius*) (Tanimoto et al., 2024). Three pathways related to nucleotide metabolism were drawn manually in PathVisio 3.3.0 (Kutmon et al., 2015), using Supplemental Figure S10A–C of Tanimoto et al. (2024) as references: the pentose phosphate pathway, pyrimidine biosynthesis, and purine and histidine biosynthesis (Table 2). Because no gene identifier system is established for this species, gene nodes were annotated with Arabidopsis thaliana gene IDs (AGI codes, Araport11 (Cheng et al., 2017)) and metabolite nodes with ChEBI IDs (Hastings et al., 2016).

Table 2: Pathway maps of *S. renifolius* created in BH26

Map	Pathway	Gene nodes linked to the table	Metabolite nodes linked to the table
FigS10A	Pentose phosphate pathway	19 / 33	4 / 13
FigS10B	Pyrimidine biosynthesis	7 / 7	2 / 8
FigS10C	Purine and histidine biosynthesis	13 / 19	1 / 15

The expression tables were built from the supplementary data of the same study. For the transcriptome (15,904 transcripts, TPM), an AGI code was given to each transcript from the best BLAST hit against Araport11 proteins; 11,438 transcripts received an AGI code. For the metabolome (93 compounds), ChEBI IDs were assigned by hand to all compounds, choosing the ID used in the reaction annotations of UniProt (Rhea) ((Bansal et al., 2022); (The UniProt Consortium, 2025)), because conversion from compound names or other IDs did not give a unique ChEBI ID. Only the samples of the thermogenic stage (Hot, florets and pith, four replicates each) were used.

To make QPX link map nodes to table rows, the maps and tables had to follow several rules that we found by reading the QPX source code: the table needs an `xref_id` column; the start of the numeric columns is given by a 0-based index (`expression_columns_index`); IDs are matched exactly, including letter case; and curved connectors are not drawn, while straight and elbow lines are. We therefore changed AGI codes to upper case, used the CHEBI: prefix for all ChEBI IDs as in WikiPathways and PathVisio, and redrew the curved edges of one map as straight lines. With these settings, clicking a node on a map filtered the heatmaps of both the transcriptome and the metabolome to that gene or compound (Figure 3). We also found that the Python side of QPX reads `xref_id` as an integer, so that string IDs such as AGI codes become null in `selected_expression_data`, although map display and heatmap filtering are not affected. This was reported to the developers.

The maps, tables and a notebook for display were published in the QPX data repository (<https://github.com/dogrunjp/qpx-data-pub>, data/Symplocarpus_renifolius), together with a record of the source of each file and every change made to it, so that the display can be reproduced with a fixed version of QPX (commit 83d27bf).

This work was carried out with an LLM coding agent (Claude Code, Anthropic). The agent was used to read the QPX source code, to review the hand-drawn maps against the source data, which found three wrong or missing ChEBI IDs, to write scripts for table conversion, and to draft documentation. All decisions on the data were made by the contributor.

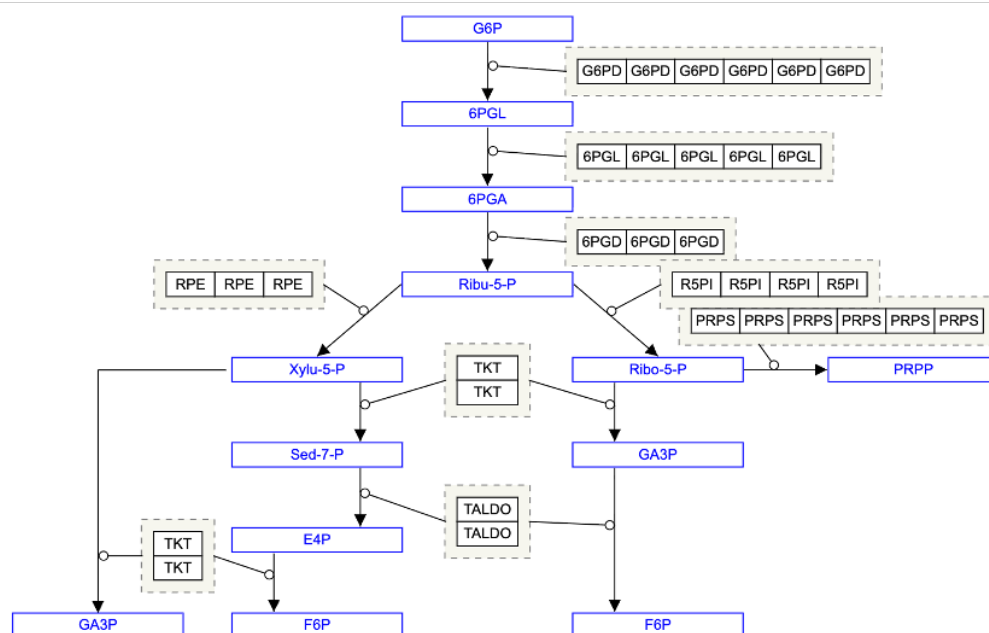


Figure 2: Pentose phosphate biosynthesis pathway

Figure 2. Pentose phosphate biosynthesis pathway of *S. renifolius* displayed in QPX.

Future work

A new methodology for constructing compound-centered functional networks was investigated by extracting relational data from PubChem Co-occurrence and enriching edge semantics with PubTator3 annotations.

We will make the assignment of ChEBI IDs reproducible. A tool that lists candidate IDs of the same compound (conjugate acids/bases and tautomers) and ranks them by their use in Rhea reactions is under development, leaving the final choice to the curator. We will also work toward automating the digitization of pathway figures into GPML, and toward handling string identifiers such as AGI codes on the Python side of QPX.

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Author contributions

Hidemasa Bono: Conceptualization, Writing - original draft, Writing – review & editing; Naoya Oec: System development, Validation, Writing – review & editing; Haruka Tanimoto: Data creation, Validation, Writing – review & editing; Tomohiro Taguchi: Data creation; Amane Kurata: Data creation; Ryo Nozu: Validation, Data creation, Writing – review & editing.

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